

Foundation (Jalapeno)

- Q1.** Solve the quadratic equation $x^2 + 5x + 6 = 0$ by factoring.
- (A) $x = -2, -3$
(B) $x = -1, -6$
(C) $x = 2, 3$
(D) $x = 1, 6$
(E) $x = -2, 3$
- Q2.** Verify if $x = 2$ is a solution to the equation $x^2 - 4x - 5 = 0$.
- (A) Yes
(B) No
(C) Maybe
(D) Not enough information
(E) Only for certain values of x
- Q3.** A basketball player scored x points in the first half and $2x + 5$ points in the second half. If the total points scored is given by the equation $x^2 + 2x - 30 = 0$, find the value of x .
- (A) $x = -6, 5$
(B) $x = -5, 6$
(C) $x = 2, 3$
(D) $x = -2, 15$
(E) $x = 5, -6$
- Q4.** A football field is x meters long. A player runs from one end to the other, and the distance is given by $x^2 - 7x - 18 = 0$. Find the length of the field.
- (A) $x = -1, 18$
(B) $x = 9, -2$
(C) $x = 2, 9$
(D) $x = -2, 9$
(E) $x = 1, 18$
- Q5.** The equation $x^2 + 4x + 4 = 0$ represents the number of wins and losses of a tennis player. Solve for x .
- (A) $x = -2$
(B) $x = 2$
(C) $x = 4$
(D) $x = -4$
(E) $x = 0$
- Q6.** A cyclist travels x miles in the first hour and $2x - 3$ miles in the second hour. The total distance traveled is given by $x^2 - x - 6 = 0$. Find the value of x .
- (A) $x = -2, 3$
(B) $x = 3, -2$
(C) $x = 2, 3$
(D) $x = -3, 2$
(E) $x = 1, 6$
- Q7.** The equation $x^2 + 2x - 15 = 0$ represents the scores of a soccer team. Solve for x .
- (A) $x = -5, 3$
(B) $x = 5, -3$
(C) $x = 3, 5$
(D) $x = -3, 5$
(E) $x = 2, 7$
- Q8.** A swimmer swims x laps in the first 30 minutes and $x + 2$ laps in the next 30 minutes. The total number of laps swum is given by $x^2 + x - 12 = 0$. Find the value of x .
- (A) $x = -4, 3$
(B) $x = 3, -4$
(C) $x = 2, 6$
(D) $x = -3, 4$
(E) $x = 4, -3$

Open Ended Questions (FRQ)

- Q9.** Solve the quadratic equation $x^2 - 3x - 10 = 0$ by factoring, and verify your solutions.
- Q10.** A volleyball team won x games and lost $x + 5$ games. The equation $x^2 + 5x - 50 = 0$ represents the total number of games played. Solve for x and verify your solutions.

Chef's Tips

- Chef's Tips
- Encourage students to factor the quadratic equations carefully.
- Remind students to verify their solutions by plugging them back into the original equation.